

1. (10 分) 设 L 是 $\{(x, y) \in \mathbb{R}^2 | x^2 + y^2 = 1, y \geq 0\}$ 。求第一型曲线积分

$$\int_L (3+x) ds \quad (1)$$

2. (10 分) 设 E 是

$$\{(x, y) \in \mathbb{R}^2 | x^2 + \frac{y^2}{4} = 1\} \quad (2)$$

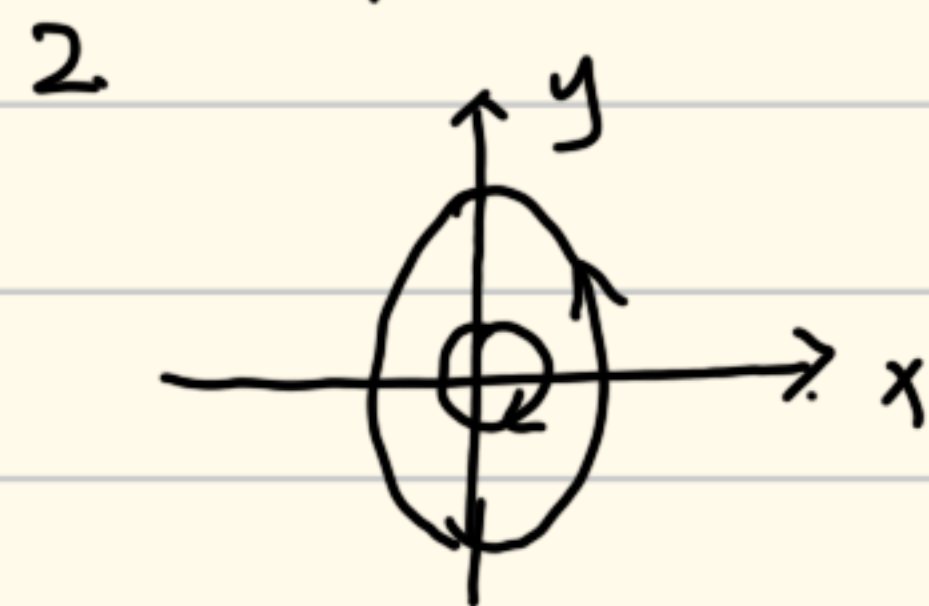
取逆时针方向。求第二型曲线积分

$$\int_E \frac{-y dx + x dy}{x^2 + y^2} \quad (3)$$



$$x = \cos \theta, y = \sin \theta,$$

$$\int_L 3 \cdot ds = 3 \cdot \pi$$



$$P = \frac{-y}{x^2+y^2} \quad Q = \frac{x}{x^2+y^2}$$

$$\frac{\partial P}{\partial y} = \frac{-x^2-y^2+y \cdot 2y}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$\frac{\partial Q}{\partial x} = \frac{x^2+y^2-2x^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

设 E_0^+ : $x^2+y^2 = \varepsilon^2$, $\varepsilon > 0$, 当 ε 充分小时, E_0 必定包含在 E 内

则 $\int_{E+E_0^+} \frac{-y dx + x dy}{x^2+y^2} = \iint_{D^+} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) d\theta = 0$

$$\begin{aligned} \therefore \int_E \frac{-y dx + x dy}{x^2+y^2} &= \int_{E_0^-} \frac{-y dx + x dy}{x^2+y^2} = \frac{1}{\varepsilon^2} \cdot \int_{E_0^-} -y dx + x dy \\ &= \frac{1}{\varepsilon^2} \cdot \iint_{D_0} 2 \cdot d\theta = \frac{2}{\varepsilon^2} \cdot 2\pi \cdot \varepsilon^2 = 2\pi \end{aligned}$$

3. (10 分) 设 D 是由直线 $y = 0, y = 2, y = x, y = x + 2$ 所围成的 $\mathbb{R}^2$ 中有界闭区域。求二重积分

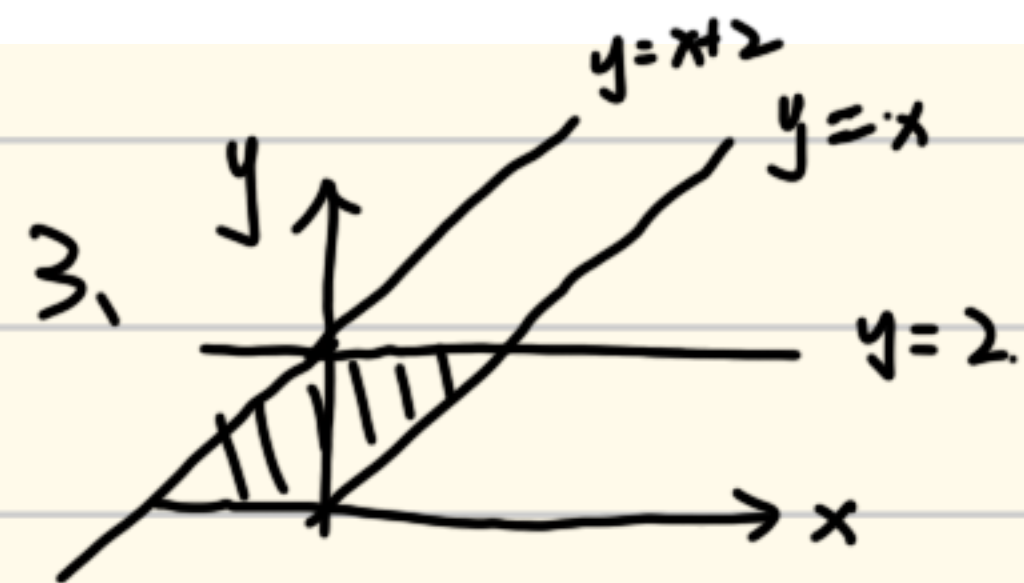
$$\iint_D \left(\frac{1}{2}x - y \right) dx dy \quad (4)$$

4. (10 分) 设 $\mathbb{R}^3$ 中曲面

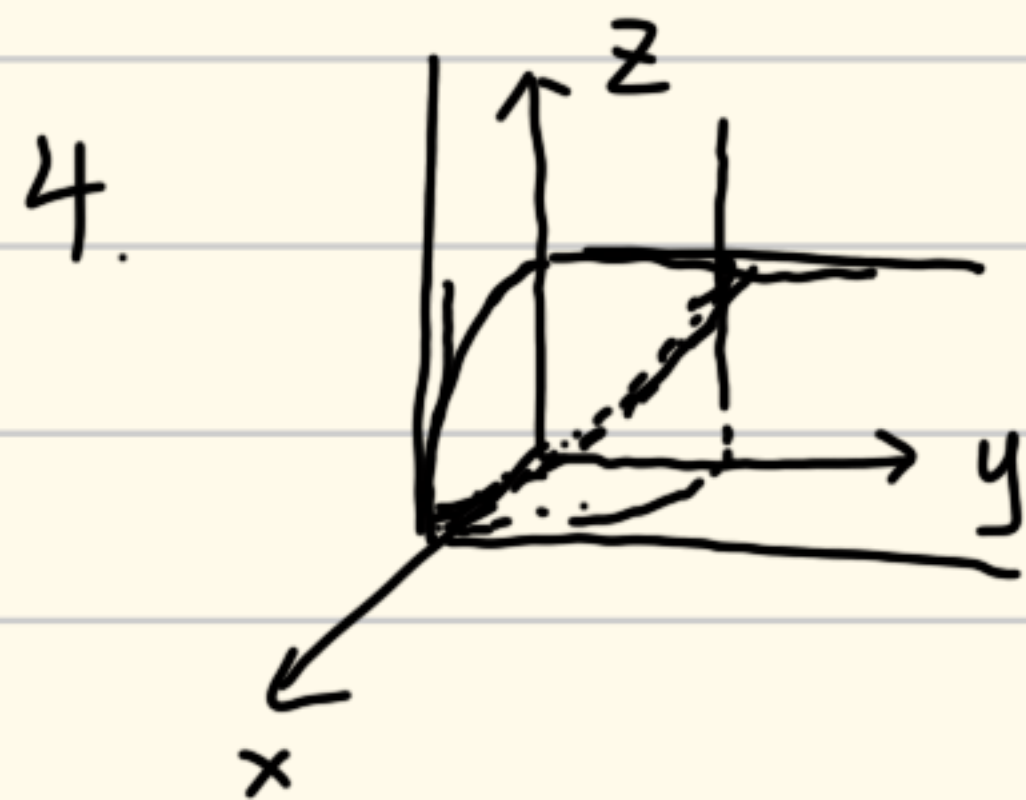
$$M = \{(x, y, z) \in \mathbb{R}^3 | x^2 + z^2 = 1, x^2 + y^2 \leq 1, x \geq 0, y \geq 0, z \geq 0\} \quad (5)$$

求第一型曲面积分

$$\iint_M x dS \quad (6)$$



$$\begin{aligned} \int_0^2 dy \int_{y-2}^y \left(\frac{1}{2}x - y \right) dx &= \int_0^2 \frac{1}{4} \cdot 2 \cdot (2y - 2) - 2y \cdot dy \\ &= \int_0^2 (y - 1 - 2y) dy \\ &= \int_0^2 (-y - 1) dy \\ &= -\frac{1}{2}y^2 - y \Big|_0^2 \\ &= -2 - 2 = -4 \end{aligned}$$



$$\begin{aligned} x &= \sqrt{1 - z^2}, \quad dS = \frac{1}{\sqrt{1 - z^2}} dy dz \\ \text{原式} &= \iint_S dy dz = \frac{1}{2} \end{aligned}$$

$$2x + 2z z_x = 0$$

$$z_x = -\frac{x}{z}$$

$$\frac{\frac{x^2}{z^2} + 1}{\frac{1}{z^2}}$$

5. (10 分) 求出一阶常微分方程初值问题 $y' = x + y^2, y(0) = 0$ 的皮卡序列的前两项 y_1, y_2

6. (10 分) 求出二阶常微分方程 $y'' - 2y' + y' = e^x$ 的通解

7. (10 分) 设 $\mathbb{R}^3$ 中有界闭区域

$$V = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + 2y^2 \leq z \leq 3 - 2x^2 - y^2\} \quad (7)$$

S^- 是 V 的边界曲面的内侧。求第二型曲面积分

$$\iint_{S^-} (x^2 + y \sin z) dy dz - (2y + z \cos x) dz dx + (-2xz + x \sin y) dx dy \quad (8)$$

5. $y_1(x) = \int_0^x (x + y^2(x)) \cdot dx$, 令 $y(x) \equiv 0$.

$$y_1(x) = \int_0^x x \, dx = \frac{1}{2} x^2.$$

$$y_2(x) = \int_0^x \left(x + \frac{1}{4} x^4 \right) dx = \frac{1}{2} x^2 + \frac{1}{20} x^5.$$

6. 特征方程 $\lambda^2 - 2\lambda + 1 = 0$, $(\lambda - 1)^2 = 0$, $\lambda_{1,2} = 1$

1 为二重根, 通解为 $(C_1 + C_2 x) \cdot e^x$

设特解为 $y = ax^2 e^x$,

$$y' = a(x^2 + 2x)e^x, \quad y'' = a(x^2 + 4x + 2) \cdot e^x$$

$$\therefore 2ae^x = e^x, \quad a = \frac{1}{2}$$

$$\therefore y^* = \frac{1}{2} x^2 e^x$$

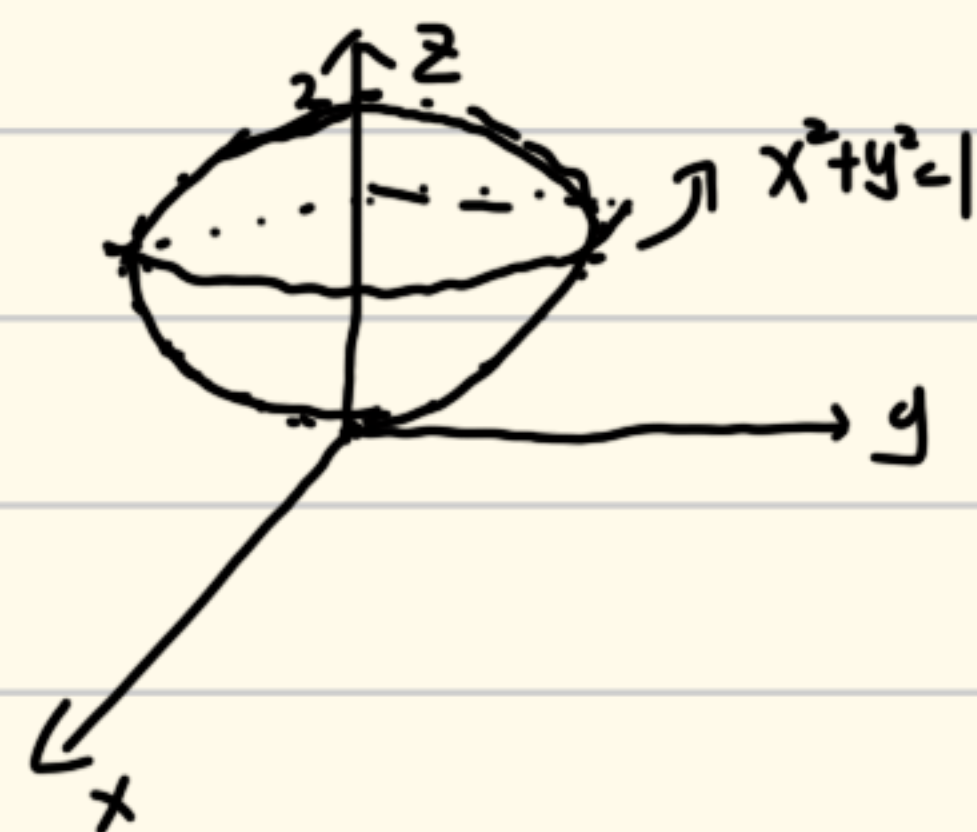
$$\therefore \text{通解为 } (C_1 + C_2 x) e^x + \frac{1}{2} x^2 e^x$$

7. (10 分) 设 $\mathbb{R}^3$ 中有界闭区域

$$V = \{(x, y, z) \in \mathbb{R}^3 | x^2 + 2y^2 \leq z \leq 3 - 2x^2 - y^2\} \quad (7)$$

S^- 是 V 的边界曲面的(内侧), 求第二型曲面积分

$$\iint_{S^-} (x^2 + y \sin z) dy dz - (2y + z \cos x) dz dx + (-2xz + x \sin y) dx dy \quad (8)$$



$$3x^2 + 3y^2 = 3, \quad x^2 + y^2 = 1$$

原式 = $\iint_{S^+} -(x^2 + y \sin z) dy dz + (2y + z \cos x) dz dx + (2xz - x \sin y) dx dy$
由高斯公式,

原式 = $\iiint_V (-2x + 2 + 2x) dV = 2 \iiint_V dV$

$$= 2 \cdot 4 \cdot \int_0^{\frac{\pi}{2}} d\theta \int_0^1 dr \cdot \int_{r^2 + r^2 \sin^2 \theta}^{3 - r^2 - r^2 \cos^2 \theta} r \cdot dz$$

$$= 8 \cdot \int_0^{\frac{\pi}{2}} d\theta \int_0^1 (3 - 3r^2) \cdot r \cdot dr$$

$$= 24 \cdot \frac{\pi}{2} \cdot \int_0^1 (1 - r^2) \cdot r \cdot dr$$

$$= 6\pi \cdot \int_0^1 (r - r^3) \cdot dr$$

$$= 6\pi \cdot \left(\frac{1}{2} - \frac{1}{4}\right)$$

$$= \frac{3}{2}\pi$$

8. (15 分) 设 r 是正实数, $f: (-r, r) \rightarrow \mathbb{R}$ 连续, $(f(0) = 0, f$ 在 0 点可导,) 对于每个 $t > 0$ 定义

$$V(t) = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + 16y^2 + \frac{z^2}{25} \leq t^2\} \quad (9)$$

$$\lim_{t \rightarrow 0} \frac{1}{t^5} \iiint_{V(t)} f\left(x^2 + 16y^2 + \frac{z^2}{25}\right) dx dy dz = \pi f'(0) \quad (10)$$

9. (15 分) 求出所有的可导函数 $f: \mathbb{R} \rightarrow \mathbb{R}$ 满足

$$f'(x) = xf(x) + x \int_0^1 tf(t) dt \quad (11)$$

$$x = r \cdot \sin \varphi \cos \theta, \quad y = \frac{1}{4} r \sin \varphi \sin \theta, \quad z = 5 r \cos \varphi$$

$$|J| = \frac{5}{4} r^2 \sin \varphi$$

$$\begin{aligned} \iiint_{V(t)} f\left(x^2 + 16y^2 + \frac{z^2}{25}\right) dx dy dz &= \iiint_{V(t)} f(r^2) \cdot \frac{5}{4} r^2 \sin \varphi d\theta d\varphi dr \\ &= \frac{5}{4} \cdot \int_0^t f(r^2) \cdot r^2 dr \int_0^{2\pi} \sin \varphi d\varphi \cdot \int_0^{\pi} d\theta \\ &= \frac{5}{4} \cdot 2\pi \cdot \int_0^t f(r^2) \cdot r^2 dr \end{aligned}$$

$$\text{所以} = \frac{5}{4} \cdot 2\pi \cdot \lim_{t \rightarrow 0} \frac{\int_0^t f(r^2) \cdot r^2 dr}{t^5} = 2\pi f'(0)$$

$$\text{只要证} \quad \lim_{t \rightarrow 0} \frac{\int_0^t f(r^2) \cdot r^2 dr}{t^5} = \frac{1}{5} f'(0)$$

$$f(0) = 0, \quad \lim_{t \rightarrow 0} \int_0^t f(r^2) \cdot r^2 dr = 0$$

$$\therefore \lim_{t \rightarrow 0} \frac{\int_0^t f(r^2) \cdot r^2 dr}{t^5} = \lim_{t \rightarrow 0} \frac{f(t^2) \cdot t^2}{5t^4} = \lim_{t \rightarrow 0} \frac{f(t^2)}{5t^2} = \frac{1}{5} \lim_{t \rightarrow 0} \frac{f(t^2) - f(0)}{t^2 - 0} = \frac{1}{5} f'(0) \quad (\text{用定义})$$

9. (15 分) 求出所有的可导函数 $f: \mathbb{R} \rightarrow \mathbb{R}$ 满足

$$f'(x) = xf(x) + x \int_0^1 tf(t)dt$$

$\int_0^1 tf(t)dt$ 本质上是常数

$$y = f(x)$$

$$\frac{dy}{dx} = xy + x \int_0^1 tf(t)dt$$

$$\frac{dy}{y + \int_0^1 tf(t)dt} = x dx$$

$$\therefore \ln |y + \int_0^1 tf(t)dt| = \frac{1}{2}x^2 + C$$

$$\therefore y + \int_0^1 tf(t)dt = C \cdot e^{\frac{1}{2}x^2}$$

$$f(x) = Ce^{\frac{1}{2}x^2} - \int_0^1 tf(t)dt = Ce^{\frac{1}{2}x^2} - k$$

$$\frac{3}{2}k = \sqrt{e} - 1, \quad k = \frac{2}{3}(\sqrt{e} - 1)$$

$$\begin{aligned} k &= \int_0^1 t \cdot (Ce^{\frac{1}{2}t^2} - k) \cdot dt \\ &= \int_0^1 (te^{\frac{1}{2}t^2} - kt) dt \\ &= e^{\frac{1}{2}t^2} - \frac{1}{2}kt^2 \Big|_0^1 \\ &= \sqrt{e} - \frac{1}{2}k - 1 \end{aligned}$$

$$f'(x) - xf(x) = x \cdot \int_0^1 tf(t)dt$$

$$\frac{dy}{dx} = xy, \quad \frac{dy}{y} = x dx, \quad \ln y = \frac{1}{2}x^2 + C$$

$$y = Ce^{\frac{1}{2}x^2}, \quad C = \varphi(x)$$

$$y = \varphi(x)e^{\frac{1}{2}x^2}, \quad y' = \varphi'(x)e^{\frac{1}{2}x^2} + \varphi(x) \cdot x \cdot e^{\frac{1}{2}x^2}$$

$$\varphi'(x)e^{\frac{1}{2}x^2} = x \cdot \int_0^1 tf(t)dt$$

$$\varphi'(x) = xe^{-\frac{1}{2}x^2} \cdot \int_0^1 tf(t)dt$$

$$k = \int_0^1 tf(t)dt$$

$$= \int_0^1 C \cdot e^{\frac{1}{2}t^2} \cdot t \cdot dt$$

$$= C_1 \int_0^1 e^{\frac{1}{2}t^2} d(\frac{1}{2}t^2)$$

$$= C_1 (\sqrt{e} - 1)$$

$\therefore$

$$f'(x) - xf(x) = C(xe^{-1})$$